

Fake News Classifier Project – Md. Al Imran

GitHub Repository: <https://github.com/rotoncsedu/fake-news-classifier>

Key Findings from EDA:

1. Fake articles are slightly longer on average
2. Both the mean and median are higher for Fake articles
3. Real articles have much greater variability

Compare the standard deviations:

- Fake: 149.5
- Real: 282.3

4. Real articles contain more extreme lengths

Preprocessing Applied:

- i. Handling missing values
- ii. Lowercasing
- iii. Punctuation removal
- iv. Tokenize
- v. Stopword removal
- vi. TF - IDF vectorization (max_features=20000)

Model Comparison:

Model	Accuracy	Precision	Recall	F1-Score
Multinomial Naive Bayes	0.97	0.97	0.97	0.97
Logistic Regression	0.99	0.99	0.99	0.99

Final Model

Model: Logistic Regression

Accuracy: 99.09%

Why this model?

Logistic Regression was selected as the final model because it achieved the best overall performance among the evaluated models. It obtained an accuracy of **99.09%** and outperformed Multinomial Naive Bayes across all evaluation metrics, including Precision, Recall, and F1-Score. Since it achieved the highest F1-Score, indicating the best balance between Precision and Recall, it was considered the most reliable model for classifying Fake and Real news articles.